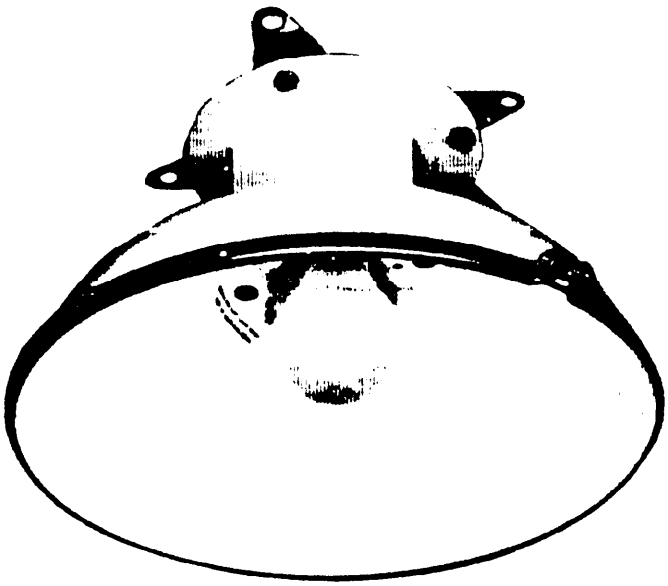
MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT, GENERAL LIGHTING

100 WATTS, 120 VOLTS, DRIPPROOF 15 DEGREES, SYMBOLS 89 AND 90.2

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH).



SYMBOL 89



SYMBOL 90.2

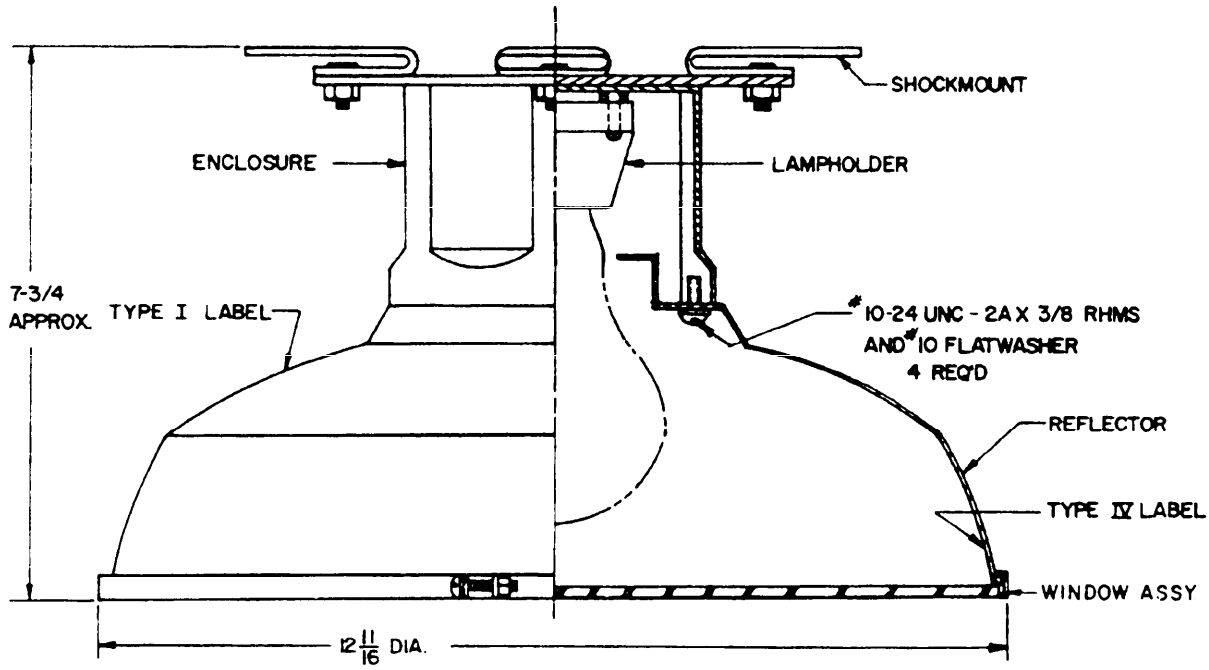
SH12031

FIGURE 1. Typical view of Symbols 89 and 90.2.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

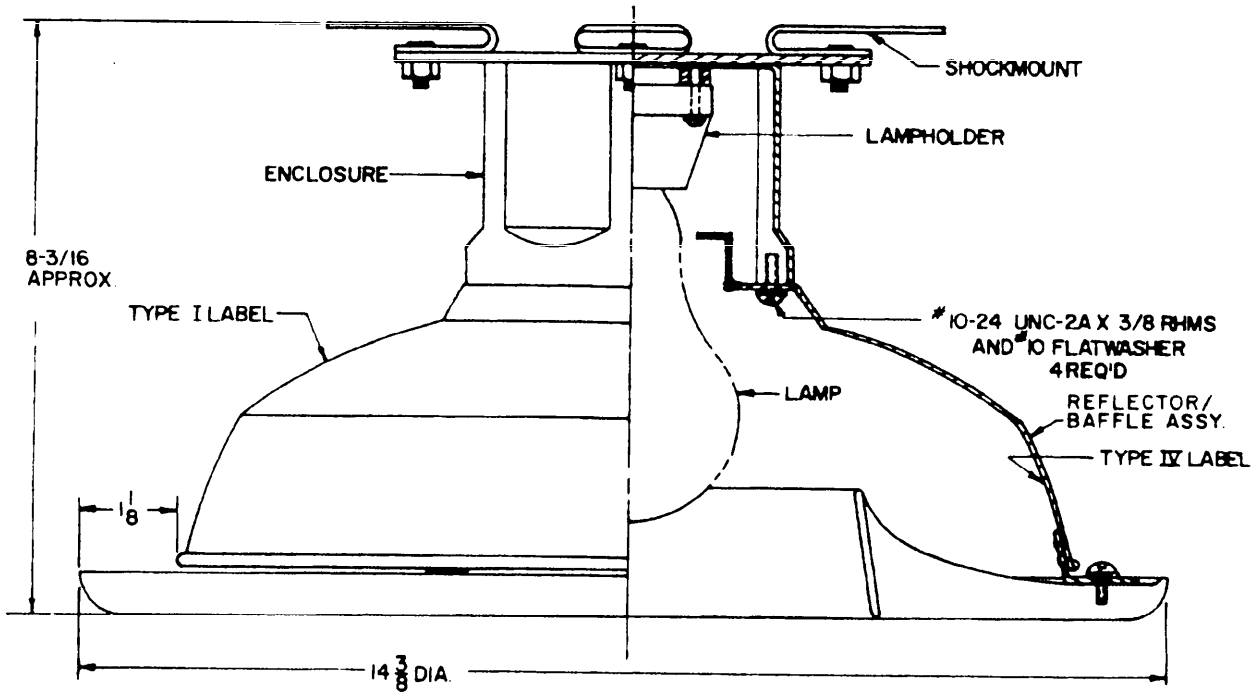
FSC 6210

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SH 87/8A

FIGURE 2. Fixture assembly, Symbol 89.



SH 87/9A

FIGURE 3. Fixture assembly, Symbol 90.2.

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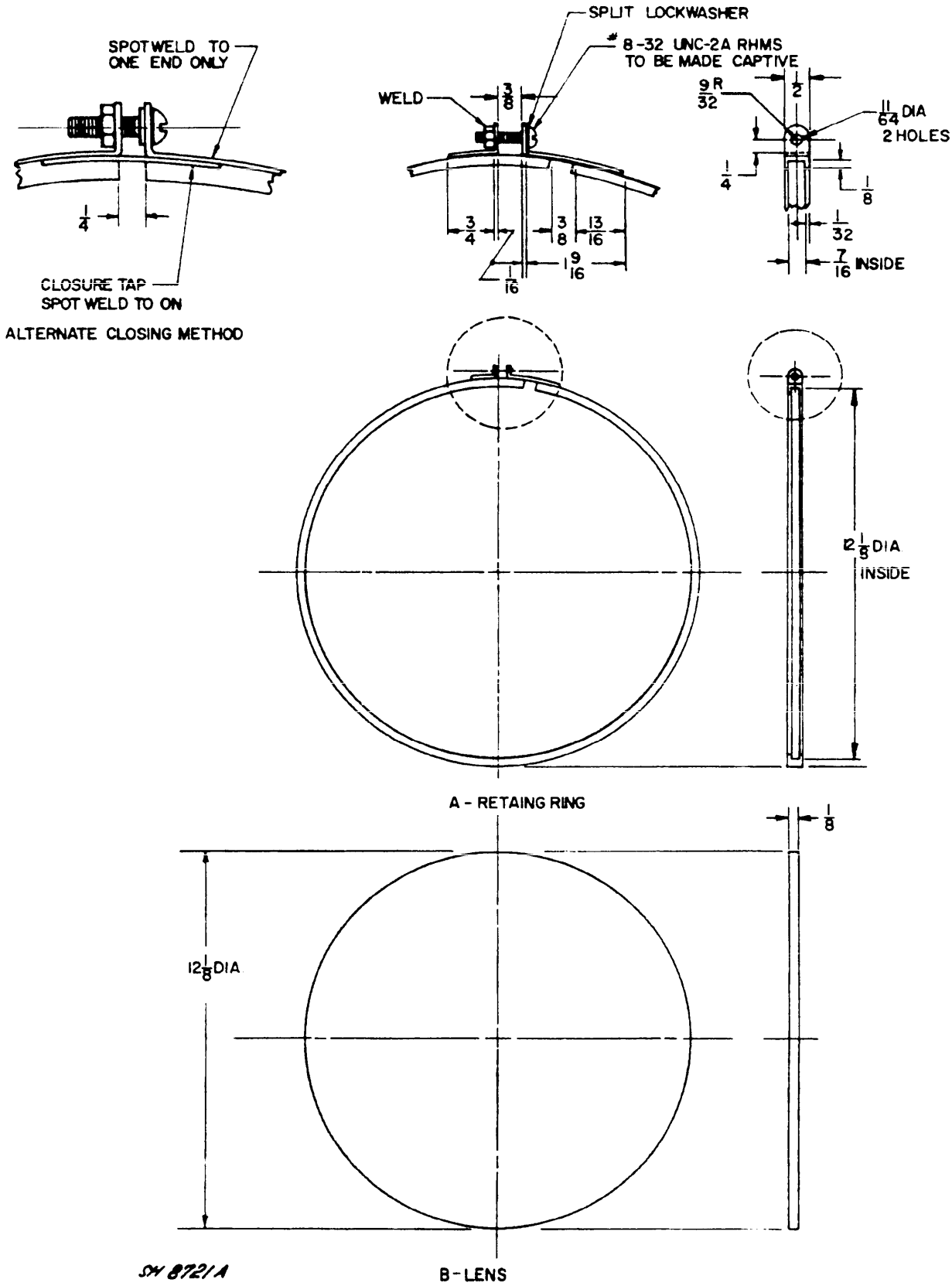


FIGURE 4. Window assembly.

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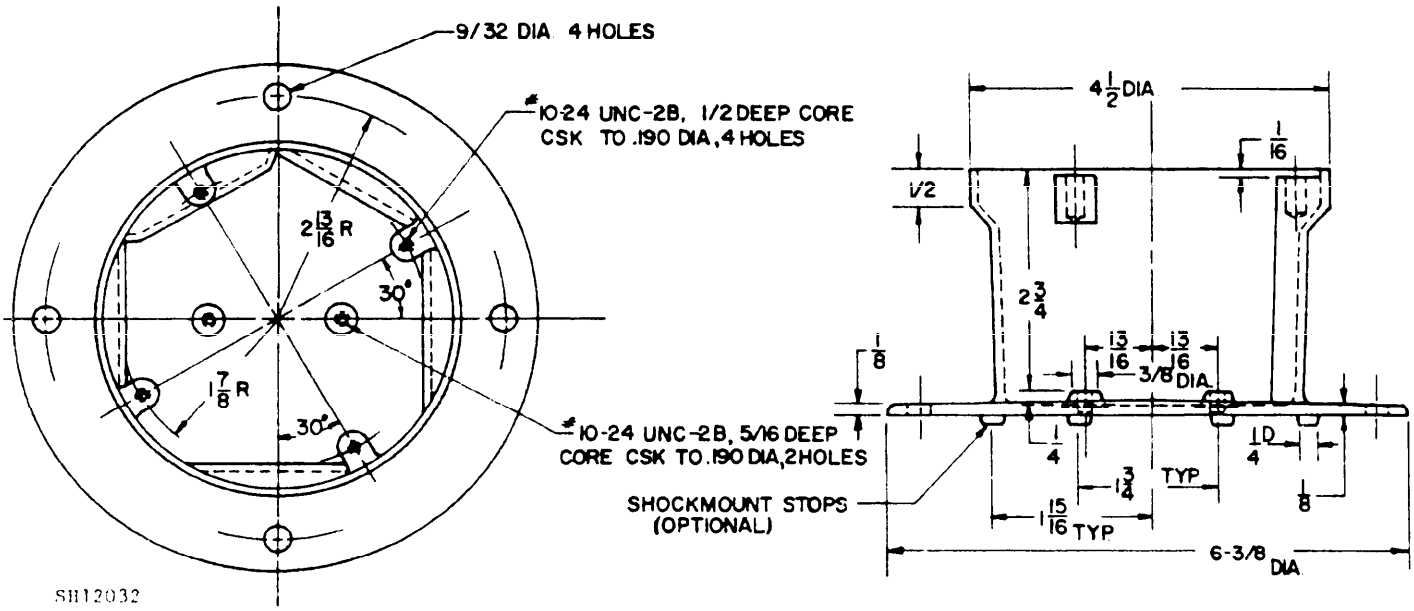


FIGURE 5. Enclosure.

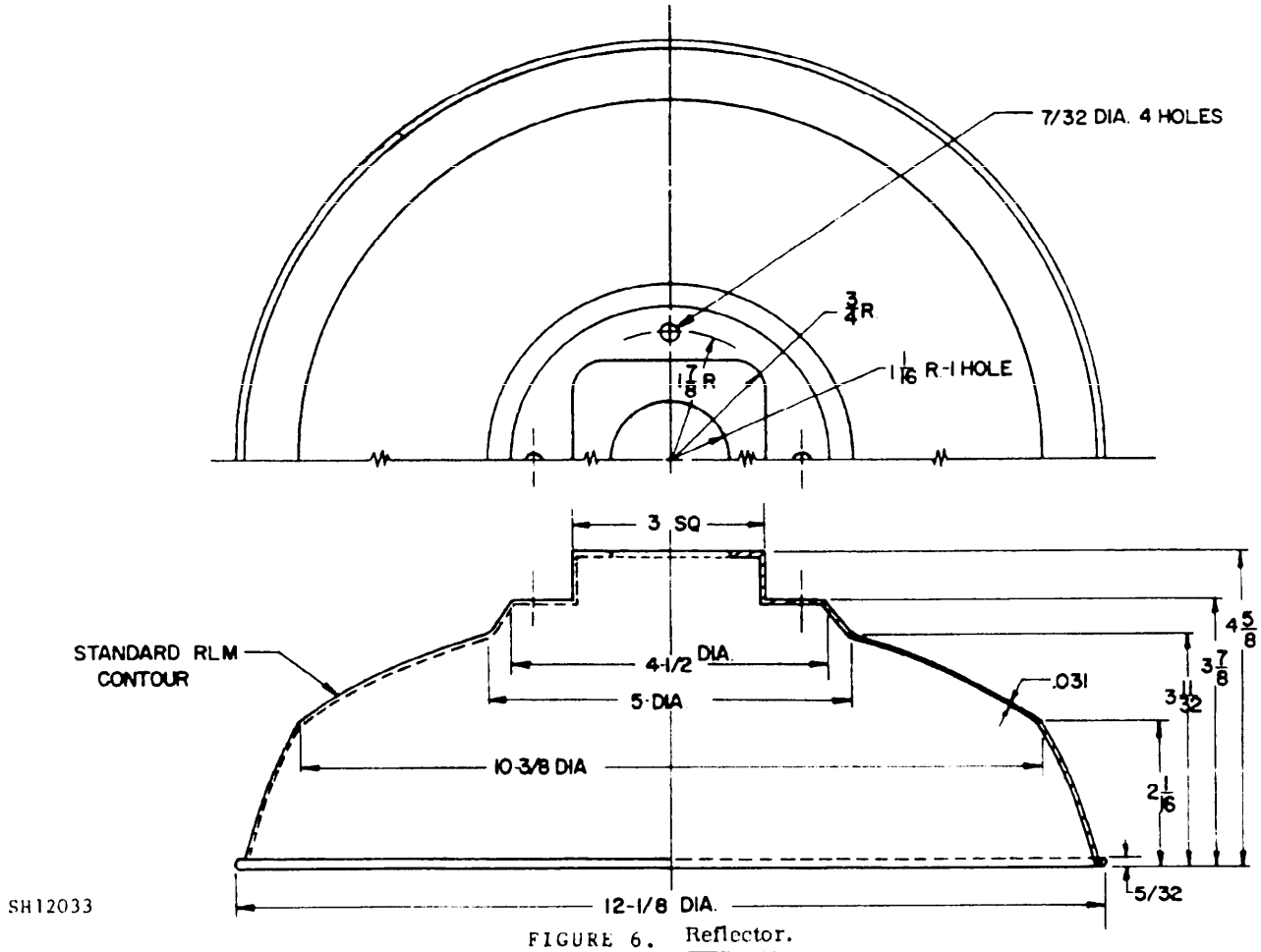


FIGURE 6. Reflector.

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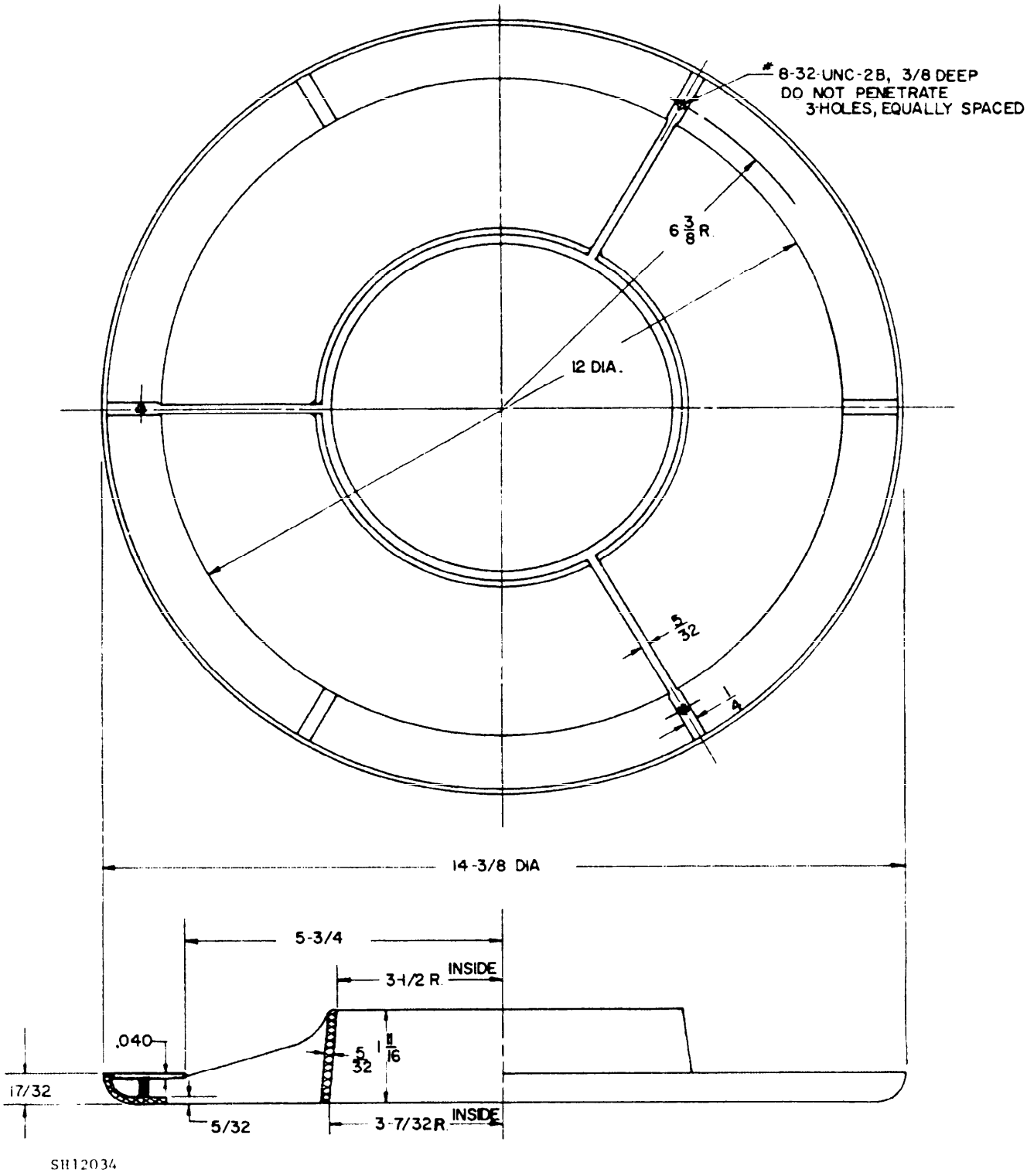


FIGURE 7. Baffle.

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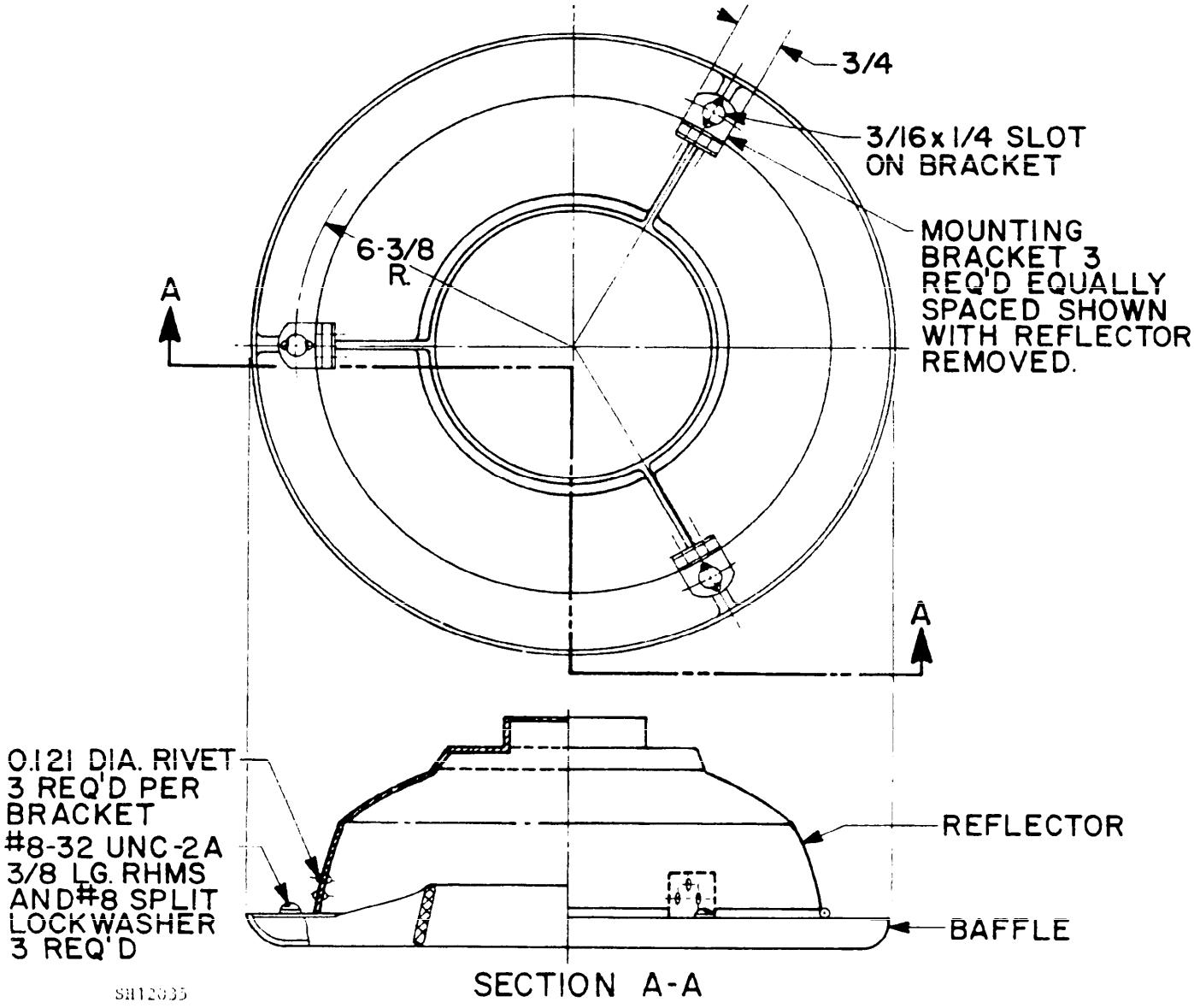
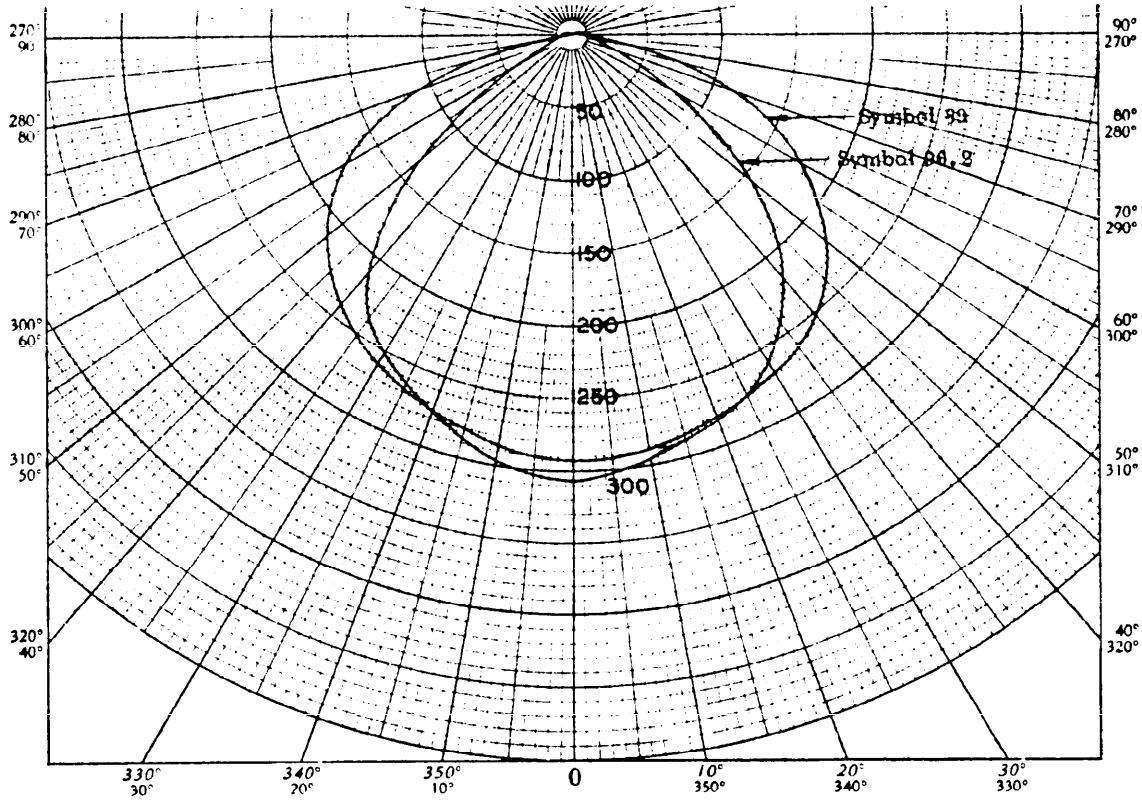


FIGURE 8. Reflector/baffle assembly.

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SH8722

ANGLE	CANDLEPOWER	
	SYM 89	SYM 90.2
0	292	301
5	290	299
10	285	292
15	279	282
20	274	273
25	272	266
30	265	256
35	259	248
40	247	218
45	236	183
50	220	133
55	204	83
60	185	60

ANGLE	CANDLEPOWER	
	SYM 89	SYM 90.2
65	158	35
70	123	24
75	38	16
80	20	10
85	10	4
90	0	0
95		
100		
105		
110		
115		
120		
125		

SH8722A

Photometric test data:

Distance: 10 feet.
Voltage: 120 volts.
Ambient temperature: 76°F.
Lamp: 100 watts, rough service conforming to MS15541-2, furnished by laboratory for test, shall be used.

FIGURE 9. Candlepower distribution curves.

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REQUIREMENTS:

CLASSIFICATION:
Type: II
Class: 2

SYMBOL NUMBERS:
89: Complete lighting fixture assembly with reflector and window assembly.
90.2: Complete lighting fixture assembly with reflector and baffle assembly.

DESIGN:
Electrical rating: 100 watts, 120 volts
Dimension and configurations: See figures 1 through 8.
Mounting dimensions: See figure 28 of MIL-F-16377.
Photometric data: See figure 9.
Enclosure assembly: Material for enclosure and reflector shall be aluminum.
Window assembly for Symbol 89 only: See figure 4. Material for lens shall be polycarbonate, clear with light transmittance exceeding 80 percent, and shall conform to the property values of L-P-393. A retainer ring as shown on figure 4 shall be provided for attaching the window to the reflector rim. All metal parts of the retainer ring shall be steel, zinc plated.
Lampholder: MIL-L-970/6, two contact.
Baffle assembly for Symbol 90.2 only: Aluminum, cast or fabricated.
Shockmounts: MIL-F-16377/41, Part Number M16377/41-009.
Enclosure: Dripproof.
Finish: All surfaces, except shockmounts and window assembly, shall receive a baked white enamel finish.
Weight:
Symbol 89: 2 pounds, 14 ounces max.
Symbol 90.2: 3 pounds, 4 ounces max.
Lamp: These fixtures shall be designed to accommodate a 100-watt, 120 volts, A-21 bulb, inside frosted, medium screw base, rough service lamp conforming to W-L-101/77. (Lamp not furnished).
Designation and marking: The following labels shall be furnished in accordance with MIL-F-16377 applied as shown in figures 2 and 3:

(a) One type I.
(b) One type IV. Apply one type IV label (as shown on figures 2 and 3), with the following information in 3/16 inch letters: "USE LAMP - NSN6240-00-246-5052 ONLY."

QUALITY ASSURANCE: Quality assurance shall be as specified in MIL-F-16377 and as follows:

First article inspection:	
Complete fixtures	See table I
Repair parts	See table II
Quality conformance:	
Comparison inspection:	Same as the first article inspection
Inspection of product for delivery:	
Examination:	See MIL-F-16377
Examination for preparation of delivery:	See MIL-F-16377
Quality conformance tests:	See MIL-F-16377

GENERAL INFORMATION:

Assignment of Military part numbers and general information is shown in table III.

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TABLE I. First article and comparison inspection for complete fixtures.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.2	Dripproof
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.4.1.1	4.8.4	Symbol 89 only
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	

TABLE II. First article and comparison inspection for repair parts.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.4.1.1	4.8.4	1/
Light output	3.7.3.7	4.8.6.3	1/ Luminous transmittance

1/ For lens of window assembly only

TABLE III. Part numbers and general information

Military Part 1/ number M16377/21-	Item Name	National Stock Number (NSN)	Remarks
89	Complete fixture, Symbol 89	6210-00-780-8502	figure 2
90.2	Complete fixture, Symbol 90.2	6210-00-548-0081	figure 3
001	Window assembly	6210-00-897-5652	figure 4
002	Reflector	6210-00-548-0220	figure 6
003	Reflector/baffle assembly	6210-00-683-5472	figure 8

1/ Formerly covered on NAVSHIPS Drawing 9000-S6401-73842 and Military Standard MS18031.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity
Navy - SH
(Project 6210-N561)

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